

User Guide - Njord SCADA



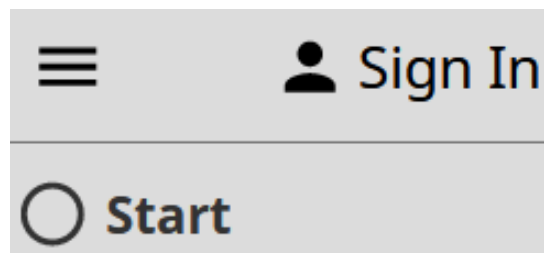
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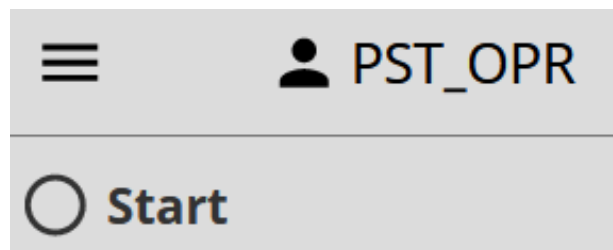
Signing in

To start using the SCADA system, users must either open the platform through a web browser or through the application “Workstation”. The platform can only be reached by devices connected to a network with access to the SCADA server. Users initially get access to viewing most of the system, but in order to modify values and access more advanced functions, they need to log into the system with a user account. Users can sign in by clicking the “Sign in” area at the top right of the top menu bar. If already logged in, the same area can be clicked to sign out.



Login-button of the SCADA system

After signing in, users are granted access to various functions based on their access level, such as more menu choices in the main menu, access to change parameter values and alarm limits, and manual operation of equipment.



Example of signed in user

Menu areas

The system has menus on 3 sides of the screen, some always visible and some expandable on demand. These menus give access to various system functions as described below.

Top menu

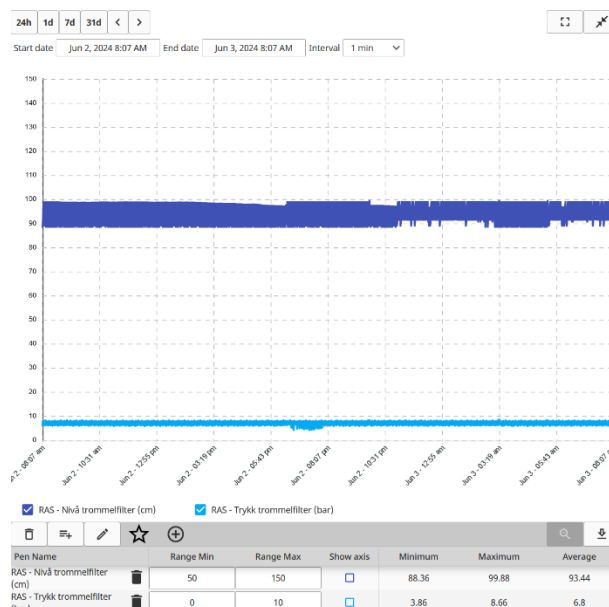
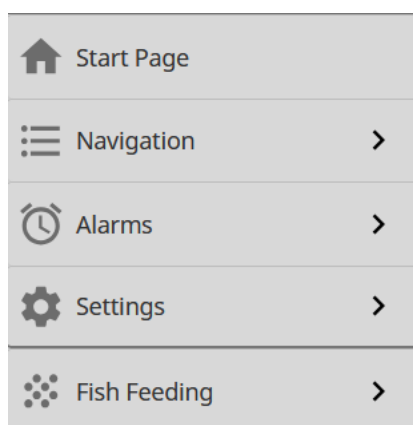
The top of the screen is always visible and gives access to some general functions as well as information about the state of the system. The left side consists of a hamburger icon that can be clicked to expand the main menu on the left side, a sign in button, and the name of the currently displayed page. The NJORD logo is in the center of the screen and can be clicked to instantly navigate to the home page of the system from anywhere (see “Navigation”). The right side consists of an alarm overview with clickable shortcuts to jump to the alarm list and some icons with quick access to the notes function (see chapter Notes), maneuver history (if authorized) and help pages.



Top menu

Left menu

The left side of the screen contains the main menu and a custom trend view (see chapter Trends). These menus are normally hidden and can be expanded by clicking their respective icons on the left or through other shortcuts such as the hamburger menu in the top menu. The main menu can be used to access a structured navigation menu for the plant and various other system functions based on the user's access level.



Main menu and custom trend

Right menus

The right side of the screen has a variety of hidden menus that change based on the current open page. They typically provide access to parameters, alarm limits and preconfigured trends related to the currently open page. They can be expanded by clicking the icons on the right side of the screen.



Example of right menus for a page

Navigation

The SCADA system consists of many menus and pages to give access to all the information and functions of a plant. The system functions are mainly accessible through the top menu and the main menu on the left side of the screen, while the process pages have several ways of navigating to give users flexible interaction and promote logical and good workflow.

The home page of the system has navigation shortcuts to the most important process pages which lets users quickly navigate to the process area they are interested in. The page can easily be reached from anywhere in the system by clicking the NJORD logo in the top menu.

Users can also use the navigation menu in the main menu on the left side of the screen to easily get access to all process pages in the plant.

Process pages usually have navigation arrows as well to indicate where process fluids come from or are being sent. If these have bold text, that indicates that they work as a shortcut and can be clicked to navigate to the relevant process page.



Example of navigation arrows, the one on the left works as a shortcut while the right is just for information

There's also a button bar at the top of process pages with shortcuts to the other screen within the same process area (e.g. same department).

Objects

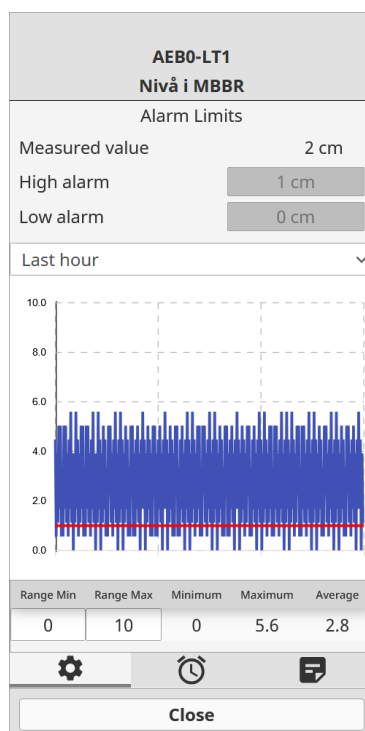
The process pages in the system use objects for electrical components to display status and give access to expanded information and manual control if it's available for the component in question. An object typically consists of a dynamic icon which can change value, color, and orientation, and gives access to more functions through a popup menu when clicked. Below is a description of the most common objects in the system. There are other components as well, but they follow the same guidelines and general functionality as the ones described.

Analog sensors

Analog sensors are shown as value fields in process pages. A red border indicates an active alarm linked to the sensor, and a red background indicates an invalid value being read (sensor fault). Clicking on the value field opens a popup with options to add the sensor value to the custom trend on the left (see chapter Trends) and usually the option to open a details page with access to more information and various sensor functions. The details menu is by default only available for sensors in our PLCs, not for other data sources such as 3rd party equipment or calculated values.



Analog sensor and popup options



Details page for an analog sensor

Motors

Motors are shown as dynamic circles in process pages, and change color and rotation based on the state of the motor. The motor will be black during standstill, change color to green during operation as well as rotate in the flow direction of the media, and blink red if there's an alarm. Clicking on the motor object opens a popup menu with options for sending the motor to the custom trend or opening a details popup for the motor. As with analog sensors, details popup is by default only available to motors controlled by our PLCs, not 3rd party vendors.



Motor object, in this case a pump. There are other variants with different graphics inside the circle.

AEB0-BM1 Blåser 1		AEB0-BM1 Blåser 1	
Control		Control	
Mode	Auto	Mode	Manual
State	Normal	Manual control	Start
Runstate	In Operation	Reference	0 hz
Speed	20 hz	State	Normal
Motor data		Runstate	In Operation
Frequency	20.0 hz	Speed	20 hz
Current	44.0 A	Motor data	
Power	0.0 kW	Frequency	20.0 hz
Motor runtime		Current	44.0 A
Today	834 min	Power	0.0 kW
Yesterday	1440 min	Motor runtime	
Total	7874 hours	Today	833 min
Motor starts		Yesterday	1440 min
Today	1 times	Total	7874 hours
Yesterday	0 times	Motor starts	
Total	11 times	Today	0 times
   		   	
Close		Close	

Details popup for a motor with variable speed control in auto/manual control

Valves

Valves are shown as dynamic objects in process pages, and change color based on the state of the valve. A gray valve indicates a hand valve or another type of valve without any electrical signals. A black valve indicates closed position, green indicates open, yellow indicates lack of position feedback and red indicates an alarm. Clicking on a non-gray valve opens a popup menu with options of sending the signal to the custom trend or opening a details popup. As with analog sensors, details popup is by default only available to valves controlled by our PLCs, not 3rd party vendors.



Valve object

FIL1-SV1
Spyleventil, trommelfilter 1

Control

Mode Auto

Status Closed

Log

Time	From		To
12/02/2025 1:53:58	Manual	→	Auto
12/02/2025 1:53:18	Open	→	Close
12/02/2025 1:53:12	Close	→	Open
07/15/2025 9:41:52	Åpne	→	Lukk
07/15/2025 9:41:43	Lukk	→	Åpne

⚙️ ⌚ 📄

Close

FIL1-SV1
Spyleventil, trommelfilter 1

Control

Mode Manual

Manual control Close

Status Closed

Log

Time	From		To
12/02/2025 1:53:18	Open	→	Close
12/02/2025 1:53:12	Close	→	Open
07/15/2025 9:41:52	Åpne	→	Lukk
07/15/2025 9:41:43	Lukk	→	Åpne

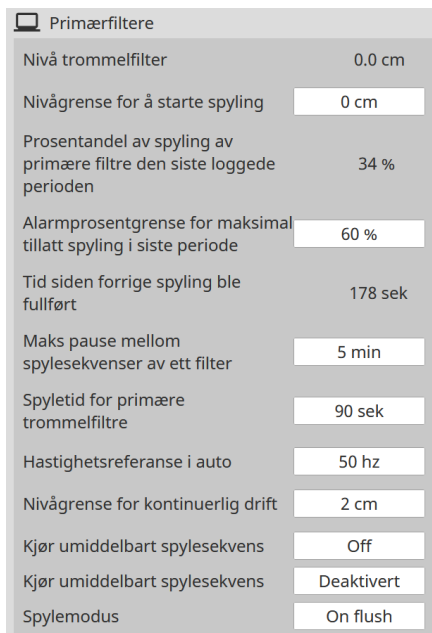
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Close

Details popup for a digital valve in auto/manual control

Value fields

Value fields are used to display various values such as parameters, alarm limits and other information in the SCADA system. There are several variants, which are roughly divided into display fields, writable fields, buttons and dropdown menus.

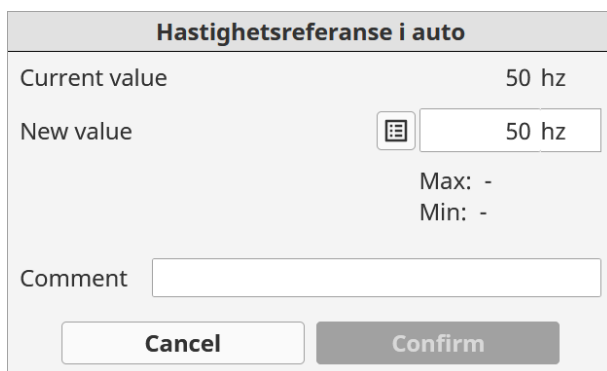


Primærfiltre	
Nivå trommelfilter	0.0 cm
Nivågrense for å starte spyling	0 cm
Prosentandel av spyling av primære filtre den siste loggede perioden	34 %
Alarmprosentgrense for maksimal tillatt spyling i siste periode	60 %
Tid siden forrige spyling ble fullført	178 sek
Maks pause mellom spylesekvenser av ett filter	5 min
Spyletid for primære trommelfiltre	90 sek
Hastighetsreferanse i auto	50 hz
Nivågrense for kontinuerlig drift	2 cm
Kjør umiddelbart spylesekvens	Off
Kjør umiddelbart spylesekvens	Deaktivert
Spylemodus	On flush

Example of value fields.

Values from display fields can be sent to the custom trend by clicking on them and selecting "Trend" from the popup menu that appears.

Writable fields will open a separate popup when clicked, which allows users with sufficient access levels to write new values to the system. Depending on the site, users may also be required to add a comment when changing the value. The popup will always have a button to toggle a quick history view and may have a trend button to send the signal to the custom trend.

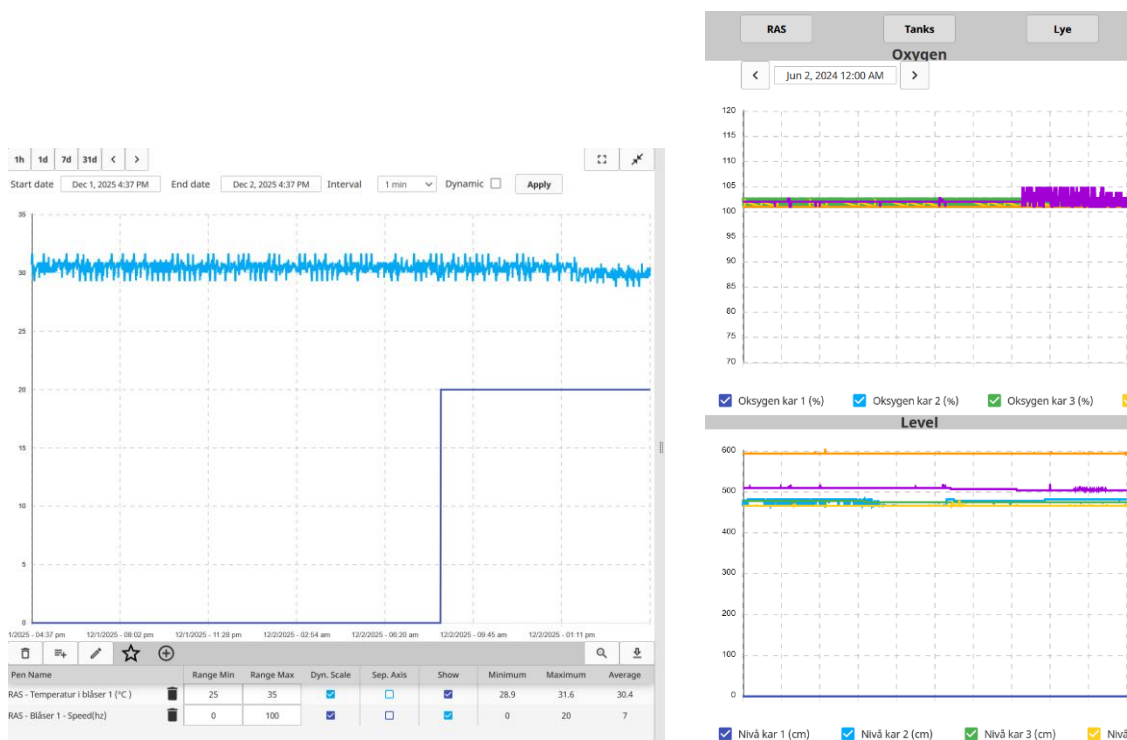


Hastighetsreferanse i auto	
Current value	50 hz
New value	50 hz
	Max: - Min: -
Comment	
Cancel Confirm	

Example of a writable field popup.

Trends

The system provides easy access to logged data through trending. There are 2 types of trends in the system: Preconfigured trends for various process areas and a custom trend where users can view any data they want.



Custom and preconfigured trends

Preconfigured trends are in the right menus of process pages where it's relevant. There are also preconfigured trends on the details popup of components such as analog sensors. They have preconfigured pens being monitored which cannot be changed by users and have limited access to modify the time period being shown. Users can zoom in the same way as described below.

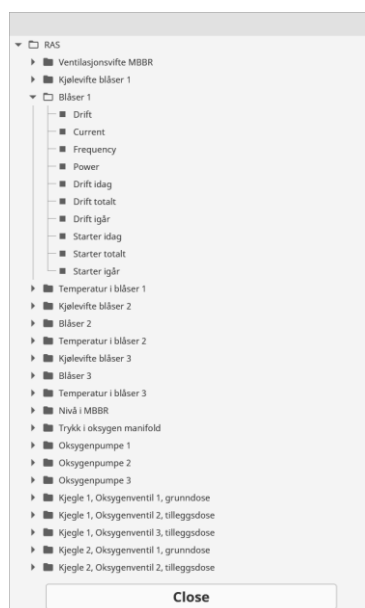
The custom trend is located on the left side of the screen and starts off empty. Users can easily add pens to the trend by clicking on components in the process images and selecting the trend option. Alternatively, users can use the signal tree browser and the pen configuration in the bottom left of the custom trend to add pens. Users can by default (must be toggled on touch devices) zoom on the trend by clicking and dragging the mouse to mark an area of the trend. When releasing the mouse button, the trend will zoom in on the selected area. Zoom can be reset by either double clicking the trend or by using the button on the bottom right under the trend.

The top of the custom trend has buttons to adjust the time period and interval data is being displayed for, with predefined buttons for quick settings. There's also a button to toggle dynamic updates of the trends. There are buttons on the righthand side to set the trend view to fullscreen and minimize the view. Below the trend is another button row with buttons to clear the trend, browse the signal tree and manage pens. The pen management also allows toggling advanced pen functions like flow estimation. Authenticated users will also be able to access favorite trends. On the righthand side are buttons for resetting zoom (+ switch

between pan/zoom for touch devices) and downloading data. There's a table at the bottom with all the configured pens which grants easy access to changing pen scaling, displaying pens in separate axes, and statistics for the time period.

Signal tree browser

The signal tree browser can be accessed by clicking the button in the bottom left below the custom trend. The browser gives access to all logged signals in the system as an expandable tree structure.



Signal tree browser

To add a signal to the custom trend, users may browse the tree until they find a desired signal and double click it to send it to the trend.

Pen configuration

The pen configuration can be accessed by clicking the button in the bottom left below the custom trend. The configuration view allows users to adjust various pen properties for the signals added to the trend and allows users to manually add new pens if they know the tag path or add custom signals. Custom signals are used to draw flat lines in the trend, typically to mark limits or other interesting breakpoints.

System pens						
Pen name	Decimals	Path	Aggregate	Accumulate flow	Color	
RAS - Temperatur i blåser 1	1	[RAS-Simulator]RAS-Sin	Last Value	Off	Dark blue	
RAS - Blåser 1 - Hastighet(hz)	0	[RAS-Simulator]RAS-Sin	Last Value	Off	Light blue	

+

Custom lines

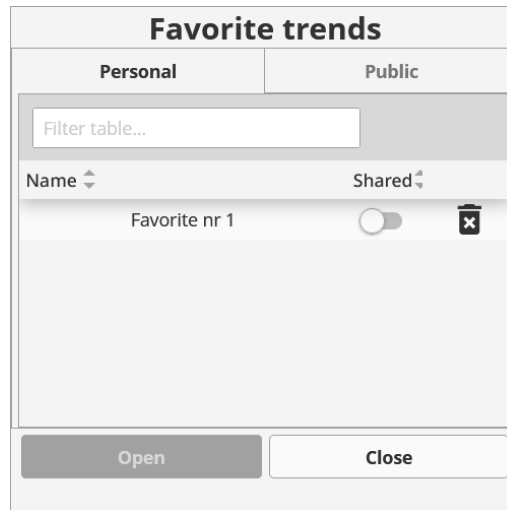
+

Cancel Apply changes

Pen configuration

Favorite trends

Favorite trends are only visible to authenticated users since they're unique for each system user. Authenticated users can create and open favorites through the buttons at the bottom left of the custom trend.



Favorite trends

By clicking the '+' icon below the trend, users can save the current pen setup with a unique name to easily be able to reload the setup later. They can also overwrite an existing favorite e.g. to add an additional pen.

By clicking the star icon below the trend, users get access to a popup menu where they can choose a favorite trend to load into the custom trend view. There are 2 types of favorites: personal and public. Personal favorites are trends saved by the user which only they can access, while public favorites are favorite trends other users have made and decided to share with all users. Personal favorites can easily be shared by clicking the toggle button in the personal list.

Reports

The SCADA system can combine logged data and generate reports for time periods. These are available from the main menu if the user has sufficient access rights.

RAS1 - Nøkkeltall							
Startdato				Sluttdato			
16-01-2023				16-02-2023			
Dato	Temperatur(°C)	pH	CO2(mg/l)	Spedevann(m³)	Daglig føring(kg)	Biomasse(kg)	Populasjon
16/01/2023	0.0	0.00	0	0	0.0	0	0
17/01/2023	0.0	0.00	0	0	0.0	0	0
18/01/2023	0.0	0.00	0	0	0.0	0	0
19/01/2023	0.0	0.00	0	0	0.0	0	0
20/01/2023	0.0	0.00	0	0	0.0	0	0
21/01/2023	0.0	0.00	0	0	0.0	0	0
22/01/2023	0.0	0.00	0	0	0.0	0	0
23/01/2023	0.0	0.00	0	0	0.0	0	0
24/01/2023	0.0	0.00	0	0	0.0	0	0
25/01/2023	0.0	0.00	0	0	0.0	0	0
26/01/2023	0.0	0.00	0	0	0.0	0	0
27/01/2023	0.0	0.00	0	0	0.0	0	0
28/01/2023	0.0	0.00	0	0	0.0	0	0
29/01/2023	0.0	0.00	0	0	0.0	0	0
30/01/2023	0.0	0.00	0	0	0.0	0	0
31/01/2023	0.0	0.00	0	0	0.0	0	0
01/02/2023	0.0	0.00	0	0	0.0	0	0
Vis beregninger							
Last ned data							

Example of a report

Reports let users freely select a time period and display data for the period in a list format. There's an expandable calculation section at the bottom and a button to download the data to export it from the system.

Maneuver history

The SCADA system logs user maneuvers, so it's possible to view historical data from changes. This history is available on a signal level.

Quick access

Users who have permission to view the maneuver history will have a quick access icon in the top row next to the note icon. Clicking it will open a popup with maneuver history from the last week, accessible from any page in the system.



Maneuver history quick access

Overview page

The system has an overview page for maneuver history where all user maneuvers registered in the system are shown with filtering options.

Maneuver History						
Start date	Select date	End date	Select date	Auto refresh	☒	
Filter table						
Time	Signal	From	To	Comment	User	
12/03/2025 8:06:48	RAS - High	1 cm	→	0 cm	Glenn Anderson	
12/02/2025 1:53:58	RAS - Spyleventil, trommelfilter 1 - Mode	Manual	→	Auto	PST Operator	
12/02/2025 1:53:18	RAS - Spyleventil, trommelfilter 1 - Manual control	Open	→	Close	PST Operator	
12/02/2025 1:53:12	RAS - Spyleventil, trommelfilter 1 - Manual control	Close	→	Open	PST Operator	
12/02/2025 1:51:08	RAS - Blåser 1 - Mode	Manual	→	Auto	PST Operator	
12/02/2025 1:50:25	RAS - Blåser 1 - Modus	Auto	→	Manuell	PST Operator	
11/27/2025 2:45:54	RAS - Parametere for beregning av for - Toggle boost feed		→	Avbryt	PST Elaut	
11/24/2025 3:00:02	Number of intervals	0	→	2	PST Elaut	
11/24/2025 2:59:57	Runtime feeder	2 sec	→	30 sec	PST Elaut	

Maneuver history overview page

Object history

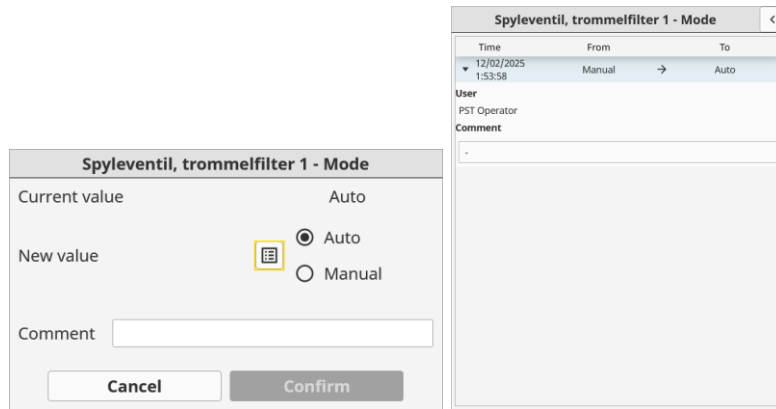
Time	From	To
12/02/2025 1:53:58	Manual	→ Auto
12/02/2025 1:53:18	Open	→ Close
12/02/2025 1:53:12	Close	→ Open

History for a maneuverable object

Maneuverable objects have a log available in the details popup where commands from users are displayed with a timestamp and the modified value.

Parameter history

Writable fields in the system have a history function which can be used to show a change log for the signal itself. It is accessed by a button on the change popup which appears when a writable field is clicked.



Time	From	To
12/02/2025 1:53:58	Manual	Auto

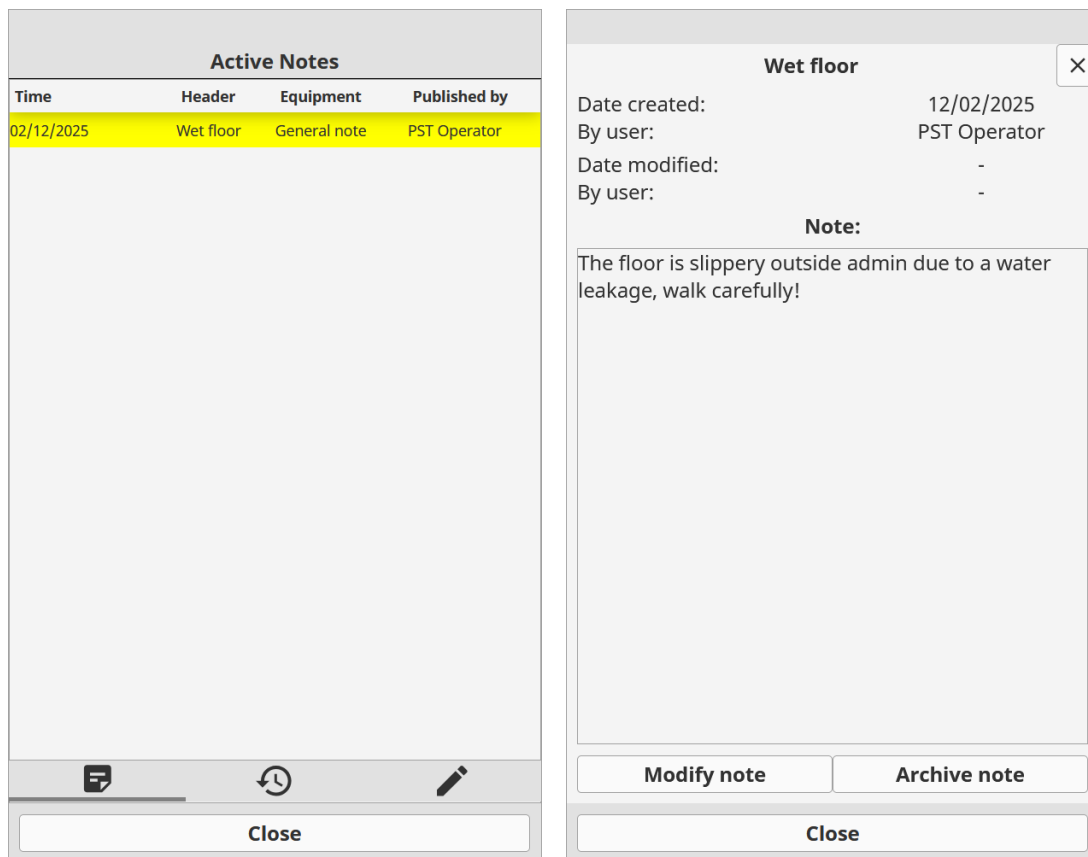
User: PST Operator

Comment:

Popup history for a writable field (history button outlined in yellow)

Notes

The SCADA system has a note function which lets users create, update and archive notes which are either general or linked to specific equipment. The system is designed to make it easy to create notes and share deviations or useful information about the plant.



Active Notes

Time	Header	Equipment	Published by
02/12/2025	Wet floor	General note	PST Operator

Wet floor [X]

Date created: 12/02/2025
 By user: PST Operator
 Date modified: -
 By user: -

Note:

The floor is slippery outside admin due to a water leakage, walk carefully!

Modify note Archive note

Close

Example of a note

General notes can be created and modified by clicking the note icon in the bottom right of the top menu (see chapter Menu areas). Clicking the icon opens an overview popup where users can create, view or modify general notes for the system.

Notes linked to equipment can be viewed and modified through the equipment details popup.

To see details and potentially modify notes in the list of active/archived notes, users can double click on the note in question to open a detailed popup with more information and edit access.

Alarms

The SCADA system handles registration and processing of alarms being monitored by the system. This is mainly equipment alarms, system alarms and other warnings/errors related to the plant. The alarms are categorized by priority and can be treated differently based on this. The alarm system consists of an alarm list to keep track of all alarm events and an alarm sender module which sends notifications to users through SMS/radio/e-mail.

Alarm rationalization

All alarms in the system start out decommissioned and need to be rationalized before they become usable alarm objects in the system. This process is known as alarm rationalization and consists of going through each individual alarm signal to determine alarm text, priority & other alarm parameters. This is done to ensure a good base for the alarm system with clear separation of critical and non-critical alarms as well as alarm texts that make sense to the site users. This is usually done during commissioning with a joint provider and customer team to ensure all considerations are taken into account. Alarms can be rationalized, re-rationalized and de-rationalized at any point by users with sufficient access.

Alarm list

The alarm list gives an overview of alarm events and the current state of the plant, as well as historical data. The alarm list can be accessed from the main menu or by clicking one of the alarm icons in the top right of the top menu. Different filters may be applied by default depending on where it was opened from.

Alarm List

Active

Disabled

Event Log

Alarm States

Filter

Filter data

Priority

Diagnostics

Low

Medium

High

Critical

State

Unacknowledged

Acknowledged

Returned

Activation Time

Area

Tag

Alarm

Priority

State

02/12/2025 08:30:21

System Alarm Notification

System

Modem 2 Lost connection to alarm modem 2

Critical

Returned

27/11/2025 14:25:05

Startfeeding MBBR

DPT2-AEB0-BM1

Blower 1 General Fault

Medium

Unacknowledged

The alarm list

At the top of the list is a set of tabs which grants easy access to switch between active alarms, disabled alarms, an alarm event log & current alarm states.

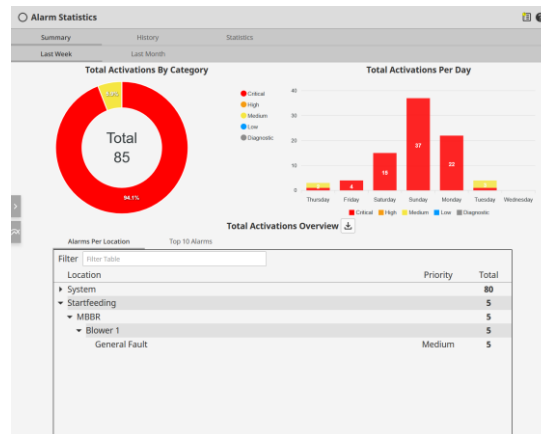
Events can be sorted by columns by clicking the arrow icons next to the column headers.

By selecting 1 or more alarms with (the entire page can be selected/deselected by using the box in the header) signed in users get access to acknowledge or block the selected alarm(s) depending on user permissions.

It's also possible to see a pre-filtered version of the alarm list on component/area level by opening a details popup for an object.

Alarm statistics

The system has a statistics page for alarms to allow analysis and comparison on different time periods and assist with identification of repeat alarms and process areas that should be adjusted.

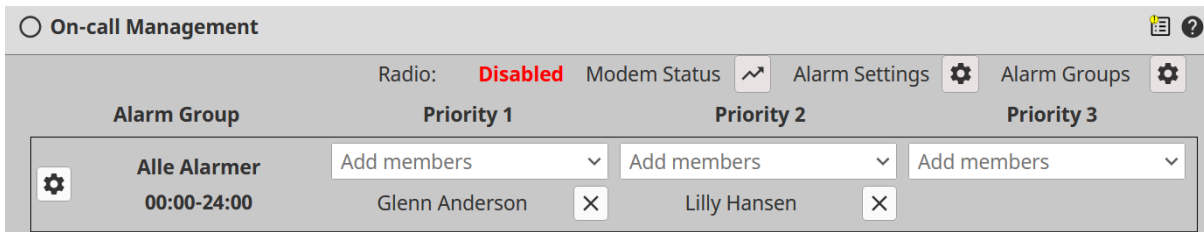


Alarm statistics page

On-call management

Overview

Users can be set up to receive alarm notification over SMS, call, radio & e-mail using the on-call management page. Users will be notified when an alarm triggers, and this will repeat with a set interval until the alarm is either acknowledged or returned. It's also possible to set up an escalating notification system where other user groups get notified if the alarm is unacknowledged for a set period. The on-call management page is accessible to users with sufficient permissions, and users may be able to access more advanced features depending on their permissions.



On-call Management

Radio: **Disabled** Modem Status: Alarm Settings: Alarm Groups:

Alarm Group	Priority 1	Priority 2	Priority 3
Alle Alarmer 00:00-24:00	Add members Glenn Anderson	Add members Lilly Hansen	Add members

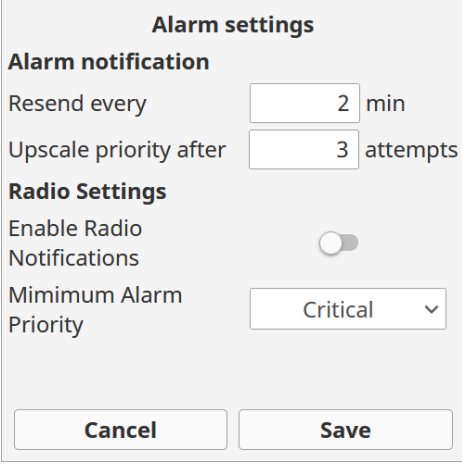
On-call roster page

The on-call management page has various status indicators and settings at the top, allowing advanced users to change how the system behaves. The rest of the page consists of a list of alarm groups where users can be set to receive notifications. Note that radio alarms are sent independently from alarm groups since it's a broadcast system with an unknown number of receivers. This also means that there's no way to select specific alarms for the radio system.

Alarm settings

This menu allows for adjustment of the following alarm notification settings:

- Resend rate: How many minutes should pass before a new notification is generated while an alarm is unhandled.
- Upscale attempts: How many notifications should be sent to the current notification priority before it's upscaled to the next priority group if applicable.
- Enable radio (if installed): Toggle radio notifications on/off.
- Minimum alarm priority for radio(if installed): Minimum required alarm priority to generate a radio notification.

A screenshot of the "Alarm settings" dialog box. The dialog has a title bar "Alarm settings". Inside, there are two sections: "Alarm notification" and "Radio Settings". Under "Alarm notification", there are two input fields: "Resend every" with a value of "2" and "min", and "Upscale priority after" with a value of "3" and "attempts". Under "Radio Settings", there is a toggle switch for "Enable Radio Notifications" which is currently turned off, and a dropdown menu for "Minimum Alarm Priority" set to "Critical". At the bottom, there are "Cancel" and "Save" buttons.

Alarm settings

Alarm notification

Resend every min

Upscale priority after attempts

Radio Settings

Enable Radio Notifications ☐

Minimum Alarm Priority

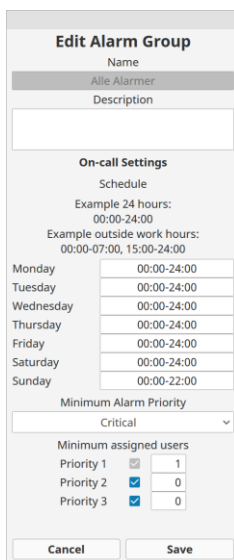
Cancel **Save**

Alarm settings

Alarm groups

It's possible to create groups of alarms for notification purposes. There needs to be at least 1 to receive notifications at all (except for radio), but there's no limit to the number, so this can be adapted to fit the plant size and user needs. Most plants have a group of all alarms for general notification purposes, and then either create more to cover different parts of the day or different plant areas. Examples can be daytime/nighttime, hatcheries, technical installations etc.

Alarm groups will have alarms assigned to them through alarm rationalization, and each group has independent settings for which parts of the day it's active for, lowest alarm priority to trigger a notification and how many notification priority groups (1-3) and minimum number of assigned users per group. The minimum number of users settings will prevent changes to the notification roster from being saved until they're met.



Edit Alarm Group	
Name	Alle Alarmer
Description	
On-call Settings	
Schedule	
Example 24 hours: 00:00-24:00	
Example outside work hours: 00:00-07:00, 15:00-24:00	
Monday	00:00-24:00
Tuesday	00:00-24:00
Wednesday	00:00-24:00
Thursday	00:00-24:00
Friday	00:00-24:00
Saturday	00:00-24:00
Sunday	00:00-22:00
Minimum Alarm Priority	
Critical	
Minimum assigned users	
Priority 1	1
Priority 2	0
Priority 3	0
Cancel Save	

Alarm group settings

User assignment & notification

Users can be assigned from priority 1-3 for alarm groups if this is enabled, where priority 1 users receive notifications immediately upon alarm activations, while higher priorities only get notified if the alarm remains unhandled for a period of time (repeat rate x upscale attempts). These settings can be managed in the alarm settings. Users will only be notified of alarms associated with their assigned alarm group with a high enough priority to warrant a notification being sent. This allows for high customization of the notification system, from all users receiving every single alarm immediately to only a few users receiving specific notifications for their assigned groups.

If the number of active and unacknowledged alarms gets too high, the system will start to send a summary notification instead of sending all the alarms individually.

Alarm messaging logs

The system has an event log for all sent notifications with a time filter to enable easy tracking and help identify issues with user notifications. It also has a basic statistics tool to compare total notifications over the last week.

Messages Audit					
Filter	Filter data	Time Period	04/22/2025 12:00am	12/04/2025 12:00am	
Event Time	Notification Type	Message	Receiver	Result	
25/11/2025 13:38:27	SMS & Call	10:41 - System - System Alarm Notification: Modem 2 Lost connection to alarm modem 2	PST Elaut	Succeeded	
25/11/2025 13:37:57	SMS & Call	10:41 - System - System Alarm Notification: Modem 2 Lost connection to alarm modem 2	PST Elaut	Succeeded	
25/11/2025 13:28:57	SMS & Call	10:41 - System - System Alarm Notification: Modem 2 Lost connection to alarm modem 2	PST Elaut	Succeeded	
24/11/2025 12:42:57	SMS & Call	22:48 - System - System Alarm Notification: Modem 1 Lost connection to alarm modem 1 10:41 - System - System Alarm Notification: Modem 2 Lost connection to alarm modem 2	PST Elaut	Succeeded	
24/11/2025 12:42:27	SMS & Call	22:48 - System - System Alarm Notification: Modem 1 Lost connection to alarm modem 1	PST Elaut	Succeeded	
15/10/2025 10:11:36	Call	14:53 - DPT2-AEB0-BM1 - Startfeeding MBBR: Blower 1 General Fault	PST Elaut	Succeeded	
15/10/2025 10:09:36	Call	14:53 - DPT2-AEB0-BM1 - Startfeeding MBBR: Blower 1 General Fault	PST Elaut	Succeeded	
15/10/2025 10:07:36	Call	14:53 - DPT2-AEB0-BM1 - Startfeeding MBBR: Blower 1 General Fault	PST Elaut	Succeeded	
15/10/2025 10:05:36	SMS	14:53 - DPT2-AEB0-BM1 - Startfeeding MBBR: Blower 1 General Fault	PST Elaut	Succeeded	
15/10/2025 10:03:21	SMS	14:53 - DPT2-AEB0-BM1 - Startfeeding MBBR: Blower 1 General Fault	PST Elaut	Succeeded	
15/10/2025 10:01:21	SMS	10:41 - System - System Alarm Notification: Modem 2 Lost connection to alarm modem 2	PST Elaut	Succeeded	

Alarm messaging logs

Test notifications and heartbeat function

The system has a page for toggling a test alarm and setting up routine notifications that can be accessed through the main menu if the user has sufficient access rights.

At the top users can write toggle a test alarm to test the notification functionality. This is useful for debugging if users have trouble receiving alarm events. At the bottom is a list where users can create routine test notifications (heartbeats) at set times, which can be useful to confirm that the notification equipment is working as intended on a daily or weekly basis.

Alarm Testing

Test alarm
Off

Heartbeat message schedule

Configured heartbeat messages

Every day at 12.00 to alarm groups []

+

Close

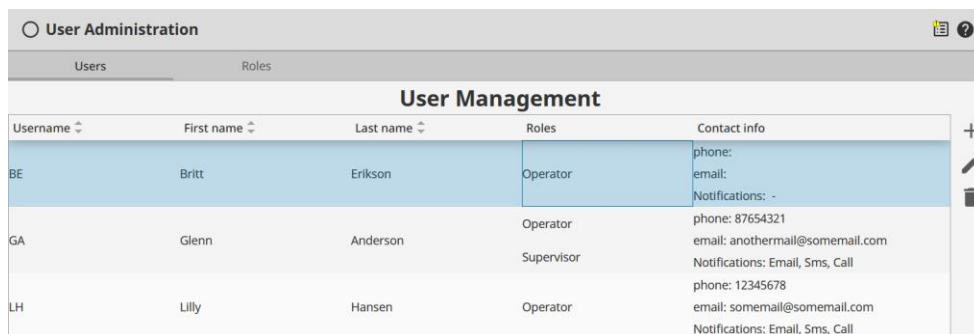
Page for alarm notification testing

User administration

User administration consists of a user management page and a roles management page. Users get assigned roles to determine what permissions they have in the SCADA system so its easy to set uniform permissions for different user levels. By default the system comes with 2 roles, but it's easy to create more to tailor the system to fit user needs.

User management

The user management page is accessible to users with sufficient permission. Depending on the permissions, users can either manage their own user, all users (including creating & deleting users) and roles. Each user has a username and password, some basic information and the possibility to set up notification information as well as role assignment. Note that users with multiple roles will combine the associated role permissions, gaining permission as long as it's assigned to at least one of its user roles.



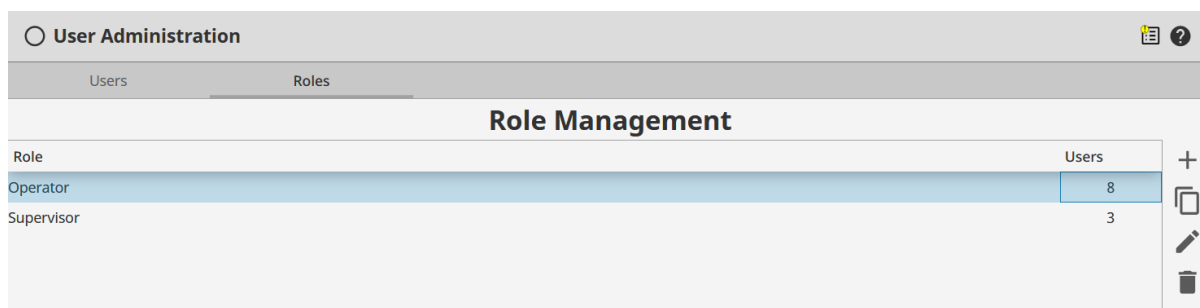
The screenshot shows the 'User Management' page within the 'User Administration' section. It features a table with columns for Username, First name, Last name, Roles, and Contact info. Three users are listed: BE (Britt Erikson, Operator), GA (Glenn Anderson, Operator and Supervisor), and LH (Lilly Hansen, Operator). The contact info column includes phone, email, and notifications for each user.

Username	First name	Last name	Roles	Contact info
BE	Britt	Erikson	Operator	phone: email: Notifications: -
GA	Glenn	Anderson	Operator Supervisor	phone: 87654321 email: anothermail@somemail.com Notifications: Email, Sms, Call
LH	Lilly	Hansen	Operator	phone: 12345678 email: somemail@somemail.com Notifications: Email, Sms, Call

User management page

Role management

Roles and their associated permissions can be managed from the Roles tab of the user management page if a user has sufficient permissions. It allows for creation, modification and deletion of roles. Each role consists of a name and a list of system permissions either being granted or denied.



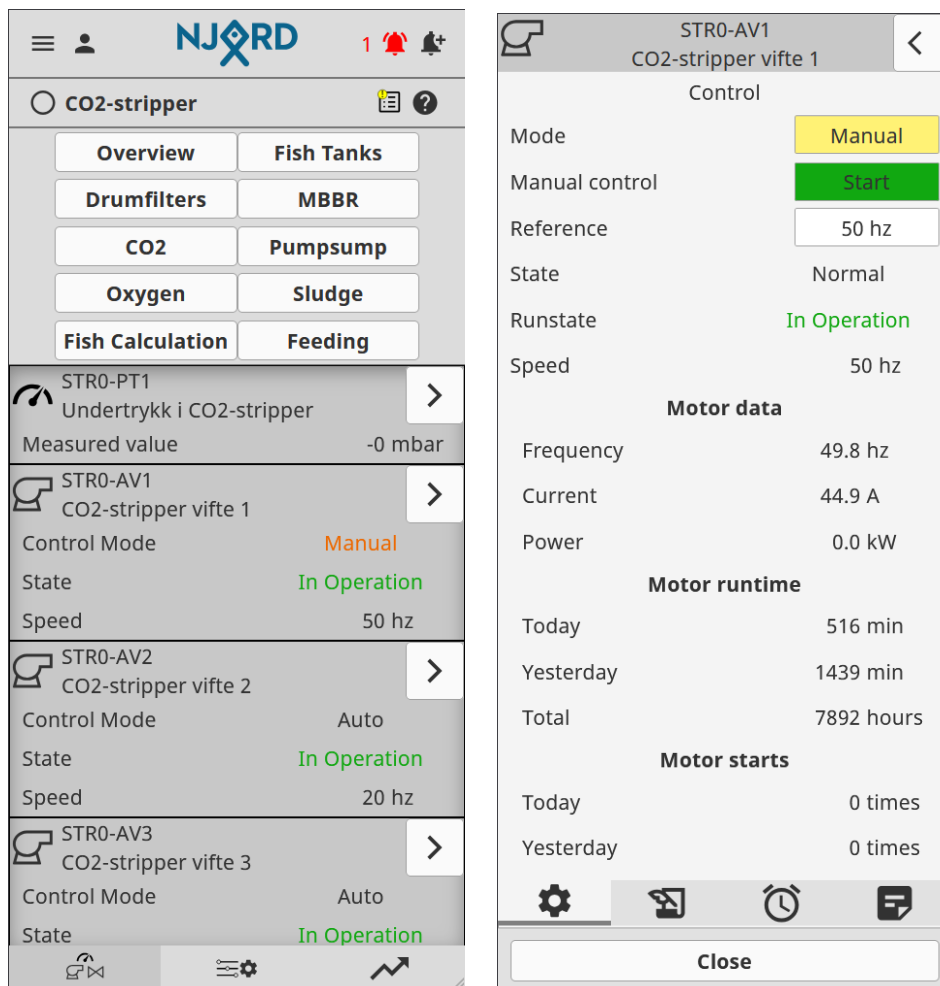
The screenshot shows the 'Role Management' page within the 'User Administration' section. It features a table with columns for Role and Users. Two roles are listed: Operator (8 users) and Supervisor (3 users). The table has a '+' icon for adding roles and a trash icon for deleting roles.

Role	Users
Operator	8
Supervisor	3

Role management page

Mobile view

The SCADA system has a compact screen solution when used with smaller screen sizes like mobile phones and tablets. It replaces the process flow view with component cards to allow for more compact display of information while still giving access to controls and relevant information.



Mobile version of a process flow view(left) & an equipment popup (right)